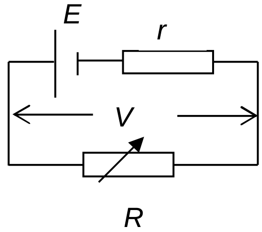
Answer: A

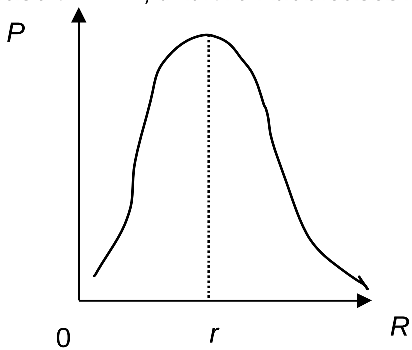
28



When $R = r$, $V = \frac{E}{2}$;

When $R = 4r$, $V'' = \frac{4E}{5}$;

Hence terminal pd increases. Power dissipated according to Maximum Power Theorem and thus it will increase till $R = r$, and then decreases as shown below.



Answer: C

29 energy supplied to wire in time $t = \frac{V^2t}{R} = \frac{V^2t}{\rho l / A} = \frac{V^2tA}{\rho l}$